Reproducible scRNA-seq Workflow (PBMC)
Technical Summary

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February 2026

1 Project Objective

This project implements a fully containerized, reproducible single-cell RNA-seq analysis pipeline using Snakemake and Docker. The workflow is demonstrated on publicly available 10x Genomics PBMC datasets (Donors 1–4).

Primary goal: demonstrate production-grade workflow engineering rather than novel biological discovery.

Key design principles:

- Deterministic execution (pinned container + pinned statistical environment)
- Version-locked dependencies (digest-pinned Docker image; `renv.lock`)
- Restart-safe modular execution (`*.done` sentinels; atomic completion markers)
- Donor-aware statistical modeling (pseudobulk DESeq2 with explicit TOST equivalence testing)

2 Architecture

Pipeline layers:

1. Python CLI wrapper (section-based execution)
2. Snakemake DAG (authoritative workflow logic)
3. Docker container (environment isolation)
4. `renv`-locked R environment (statistical layer)

Execution flow:

```
FASTQ → QC → (optional trimming) → STARsolo →
Seurat QC/Clustering → Pseudobulk DE →
Equivalence testing (TOST) → Pathway enrichment →
```

Each stage produces atomic `*.done` sentinel files to ensure restart safety and auditability.

3 Reproducibility Strategy

Containerization

- Base image: `rocker/r-ver:4.3.3`
- Digest-pinned release image published to GHCR
- All execution occurs inside Docker (host-independent runtime)

R Environment Control

- `renv.lock` version-locked
- CRAN snapshot pinned (2024-01-01)
- Bioconductor 3.18 pinned
- Seurat / SeuratObject vendored locally (installed from local tarballs)

This eliminates upstream dependency drift and reduces long-term reproducibility risk.

4 Upstream Processing

Data

10x Genomics GEM-X 3′ v4 PBMC libraries (Donors 1–4).

Quality Control

FastQC + MultiQC.
Findings:

- High base quality across donors
- No pervasive adapter contamination
- Trimming unnecessary (optional branch retained for pathological runs)

Alignment

Aligner: STARsolo v2.7.11b
Reference: GRCh38 + GENCODE v45
Chemistry: 16 bp cell barcode + 12 bp UMI
Alignment summary:
Metrics are consistent with high-quality 10x 3′ scRNA-seq data.

Donor	Unique (%)	Multi (%)
Donor1	71.9	20.2
Donor2	72.4	19.3
Donor3	72.4	19.8
Donor4	68.8	21.7

5 Downstream Analysis

Seurat Processing

- `LogNormalize` (scale factor 10,000)
- 2000 HVGs (VST)
- PCA (30 PCs)
- Louvain clustering (resolution 0.3)
- Marker-based annotation via module scores

Major PBMC lineages are recovered consistently across donors. Cell-type annotation prioritizes transparency over automated reference mapping; while this approach reliably recovers major PBMC lineages, finer subtype resolution or ambiguous populations may benefit from reference-based annotation methods in future workflow iterations.

Pseudobulk Differential Expression (DESeq2)

Counts are aggregated per donor × immune group (donor treated as the experimental unit).

Model:

$$\sim \texttt{donor} + \texttt{group}$$

Tool: DESeq2 v1.42.1

Strict marker definition:

$$\texttt{padj} < 10^{-10}, \quad |\log_2 \mathrm{FC}| > 3$$

Contrasts:

- T-like vs B-like
- T-like vs Mono-like
- B-like vs Mono-like

Equivalence Testing (TOST)

Margin:

$$\delta = 0.75$$

Conserved genes:

$$\texttt{padj}_{equiv} < 0.05 \quad \text{and} \quad |\log_2 \mathrm{FC}| < \delta$$

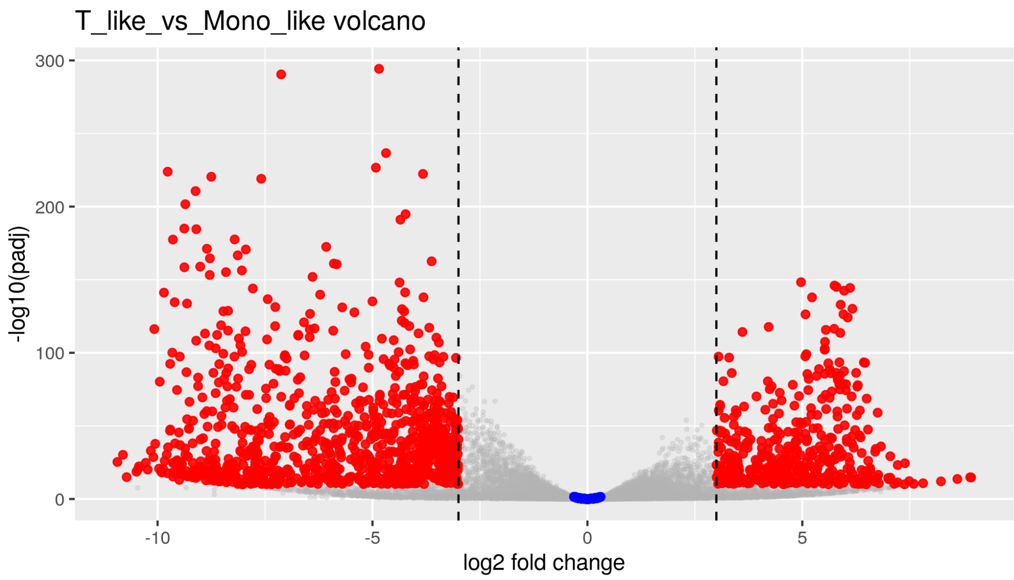

Figure 1: Representative pseudobulk DE contrast (T-like vs Mono-like). Strict markers occupy the distribution tails while near-zero effects concentrate around the origin.

This separates true small-effect similarity from statistical non-significance, yielding a contrast-specific marker set alongside a shared "core" program.

6 Pathway Enrichment

- GSEA (`fgsea`)
- ORA (`clusterProfiler`)
- MSigDB Hallmark + C7

Markers are enriched for inflammatory and immune activation programs (e.g. TNFα/NFκB and interferon signaling), while conserved genes emphasize shared metabolic and proliferative programs (e.g. oxidative phosphorylation, MYC targets, cell cycle).

7 Consensus Co-expression Networks (Muumi)

Constructed separately for the following cell-type strata (defined in `scripts/celltype_sets.R`):

- CD4 T cells
- B cells
- CD14 monocytes

Method overview:

- Donor-stratified analysis using annotated Seurat objects (one RDS per donor). Cell labels are taken from `cell_type_cluster_majority`. Donor/stratum combinations with < 200 cells are skipped.

- Random metacell aggregation ($m = 20$; seed 42) is performed using `linear_then_log` aggregation (`expm1` pooling followed by `log1p`). Genes are filtered by detection fraction ≥ 0.05.
- Highly variable genes are selected per donor (`FindVariableFeatures`, $n \leq 4000$) and intersected with the filtered metacell matrix.
- Networks are inferred with **Muumi** using the MINET ensemble (`clr` algorithm; Spearman estimator; no discretization). Ranked edges are converted to weights ($w = 1/\text{rank}$).
- Per-donor networks are sparsified using per-gene top-k neighbors ($k = 75$) and restricted to the largest connected component.
- Cross-donor consensus retains edges present in ≥ 2 donors; consensus weight is the median donor weight. The final network is restricted to its LCC.

Modules and annotations:

- Modules are detected using Louvain clustering via Muumi.
- Nodes are annotated with degree, betweenness, DEG/TOST markers, and curated PBMC markers.
- Module enrichment is computed using ORA against Hallmark and C7 gene sets, with additional Reactome enrichment via Muumi.

8 What This Demonstrates

- Production-grade Snakemake-style outputs (per-donor and consensus artifacts; restart-safe with explicit file outputs)
- Containerized statistical reproducibility (`renv` + digest-pinned Docker environment)
- Donor-aware construction of replicate-stable networks (support filtering across donors rather than single-sample inference)
- Scalable GRN/co-expression inference using **Muumi** with MINET backends (`clr`) and rank-based ensemble semantics
- Transparent sparsification and consensus rules (per-gene top-k + donor support threshold + median weight aggregation)
- Integrated module-level biological validation via GMT ORA (Hallmark/C7) and Reactome enrichment reporting

9 Data and Results Availability

Representative outputs, the reproducible toy data bundle, and versioned workflow artifacts are archived on Zenodo. All versions are accessible via the concept DOI:

`https://doi.org/10.5281/zenodo.19021322`

This DOI resolves to the latest release and provides access to all historical versions.

10 Conclusion

This workflow provides a complete, auditable, and reproducible scRNA-seq analysis system from raw FASTQs to donor-aware inference and consensus network construction.

The primary contribution is engineering rigor: reproducibility, modular execution, statistical transparency, and cross-donor validation are treated as first-class design constraints.

Full technical documentation and extended results are available in the accompanying report and repository.

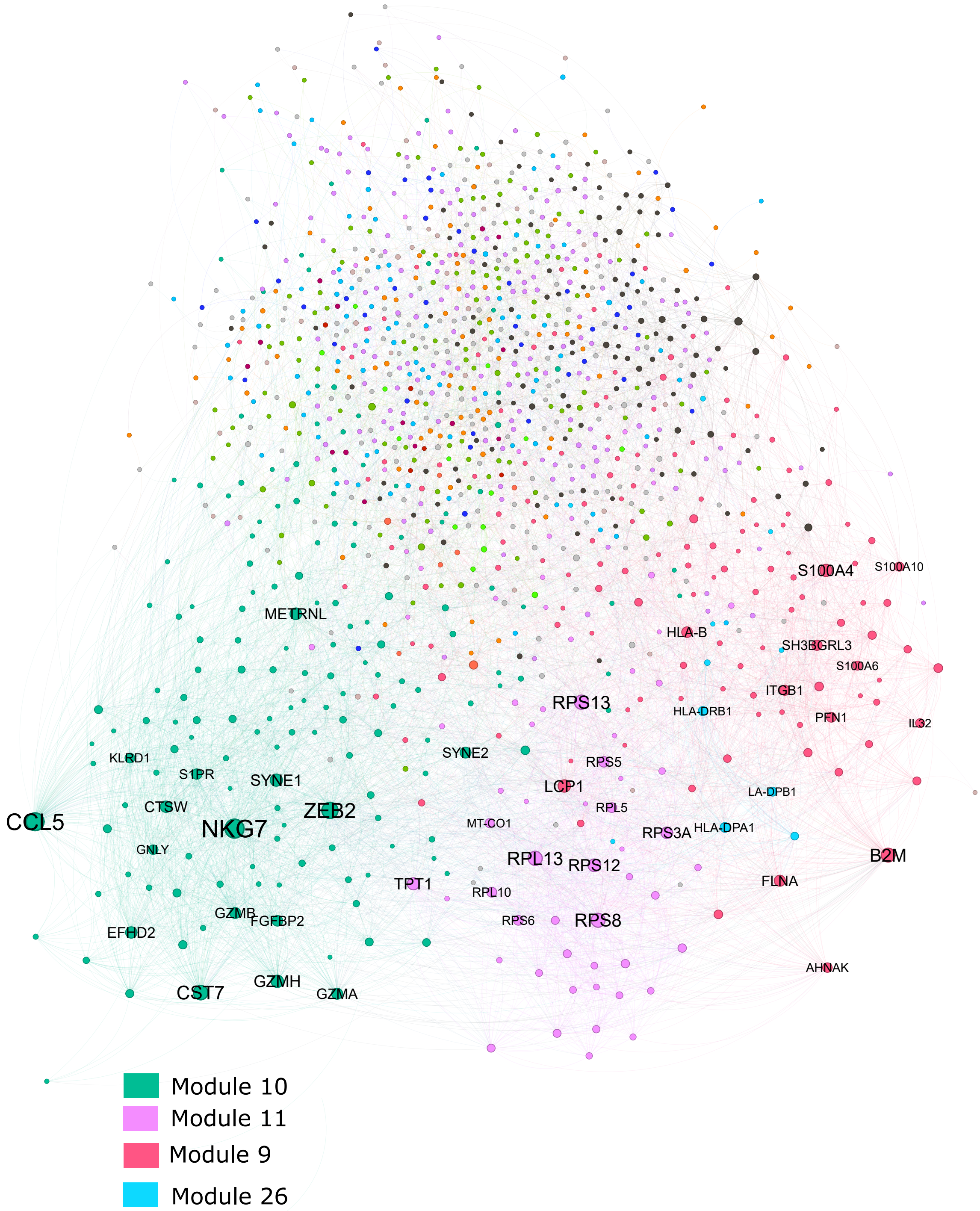

Figure 2: Example Muumi consensus network (CD4 T cells) colored by Louvain module. Edges retained in ≥ 2 donors represent replicate-stable co-expression structure used for module enrichment.